Industrial Power Management Company

UNITY ENERGY

intelligent engineered energy

Prepared for;

Foster Farms

Cherry Ave Facility

Fresno, California 93706

Energy Field Studied;

T15 — FILLET

1500kVA Transformer

(1804amps @ 480volts)

Report Topic;

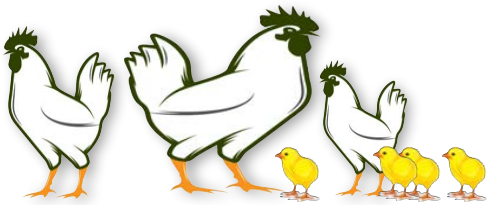
SATURATION DISCOVERY

Active Energy Field Data; UNITY METER SN: **AN53110845**

eSTREAM Images c.240404

15d@1hr, 15d@15min, + Voltage, 7day@1min,
2day@1sec, 1day@1sec, + Voltage

EXAMINE ▽ HARVEST △ REFINE ▽ MAP △ EXCHANGE ▽ OPTIMIZE



intelligentrenewable**sustainability**

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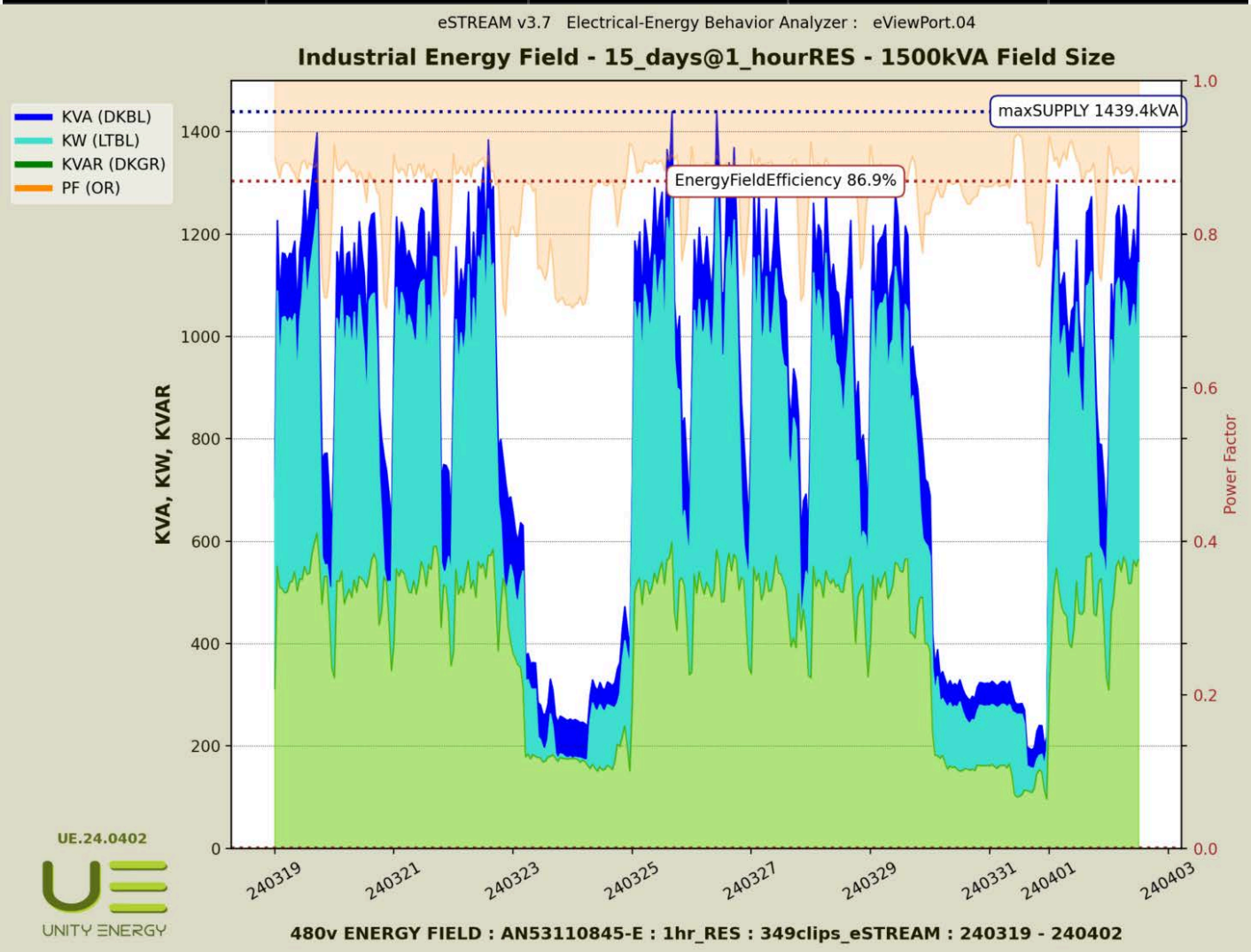
T15 Fillet -- 1500kVA Transformer -- Saturation Review

Resolution Deception , 1 hour RES, No Problems

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
93 homes or 239 EVs (1.26kW/hr) (powered continuously) (.35kW/hr)	1301.3 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reaction losses)	RECOVERABLE CO2e
113.4 kVA (unconsumed power)	862.7 kVA 621,137.8 kVA /mo	749.3 kW 539,499.4 kW /mo	409.6 kVAR 294,925.6 kVAR /mo	97.5 LBS 387.5 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
473.0 VOLTS 493.9 MAX 456.3 MIN	1037.7 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	1.6 % LIMIT 1439.4 kVA MAX	86.9 % PF (power factor)



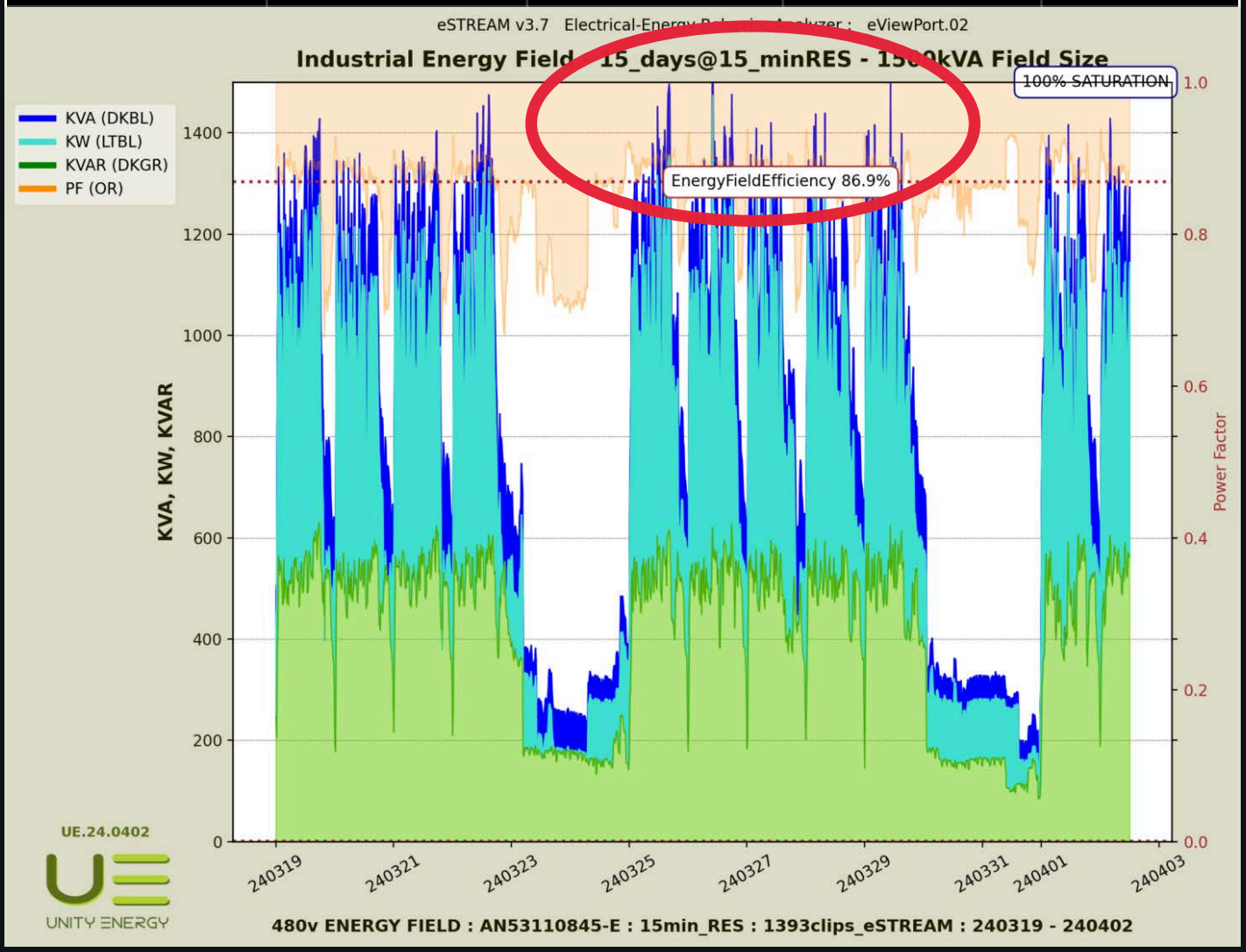
T15 Fillet -- 1500kVA Transformer -- Saturation Review

15minUtility Resolution - Only Few Saturation Points

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
93 homes or 239 EVs (1.26kW/hr) (powered continuously) (.35kW/hr)	1474.7 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reaction losses)	RECOVERABLE CO2e
113.3 kVA (unconsumed power)	861.8 kVA 620,489.6 kVA /mo	748.5 kW 538,904.4 kW /mo	409.3 kVAR 294,693.1 kVAR /mo	97.4 LBS 387.2 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
473.0 VOLTS 494.3 MAX 454.3 MIN	1036.6 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	-10.5 % LIMIT 1616.4 kVA MAX	86.9 % PF (power factor)



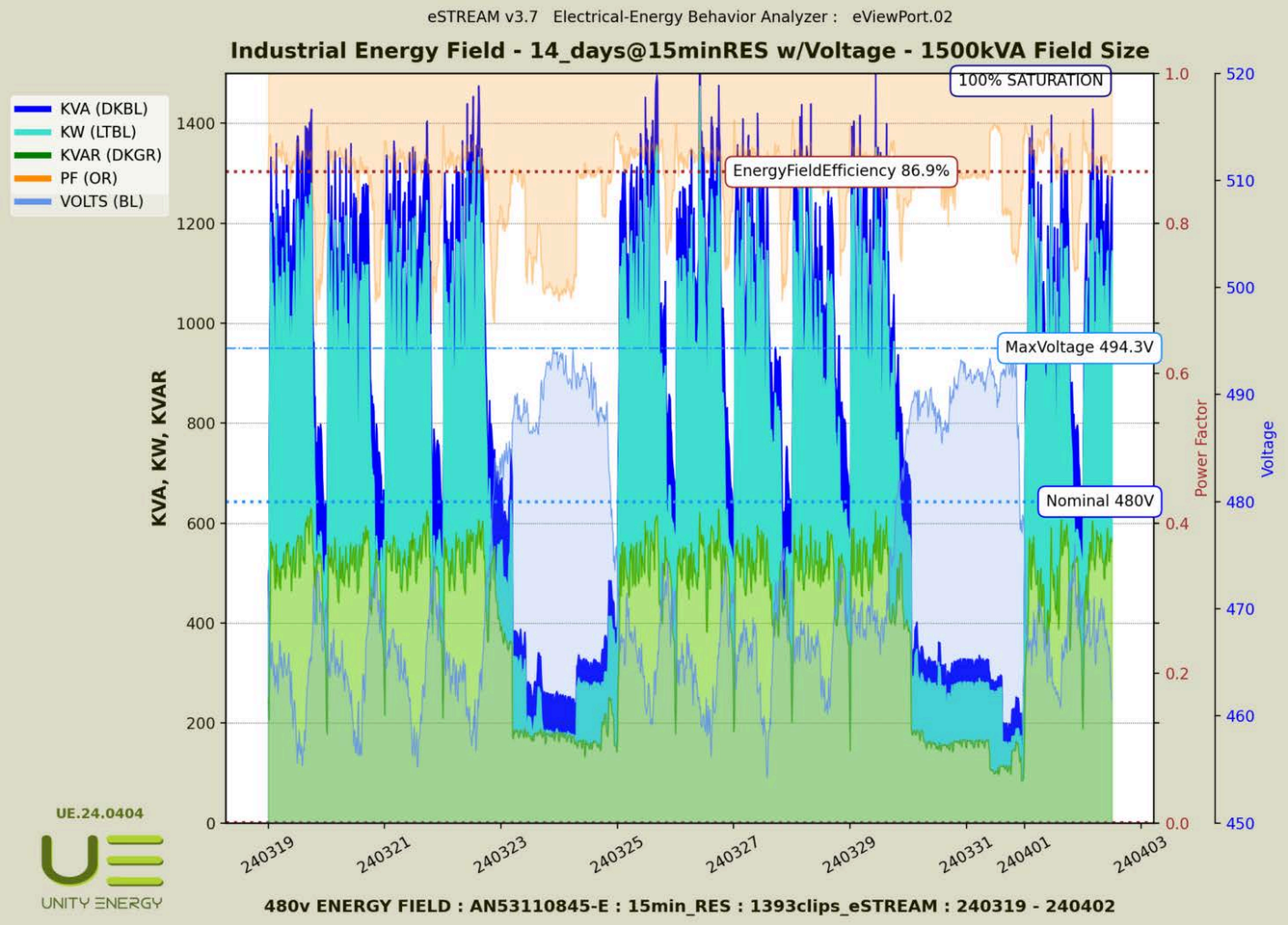
T15 Fillet -- 1500kVA Transformer -- Saturation Review

15minUtility Resolution - w/ Voltage

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
93 homes or 239 EVs (1.26kW/hr) (powered continuously) (.35kW/hr)	1474.7 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reaction losses)	RECOVERABLE CO2e
113.3 kVA (unconsumed power)	861.8 kVA 620,489.6 kVA /mo	748.5 kW 538,904.4 kW /mo	409.3 kVAR 294,693.1 kVAR /mo	97.4 LBS 387.2 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
473.0 VOLTS 494.3 MAX 454.3 MIN	1036.6 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	-10.5 % LIMIT 1616.4 kVA MAX	86.9 % PF (power factor)



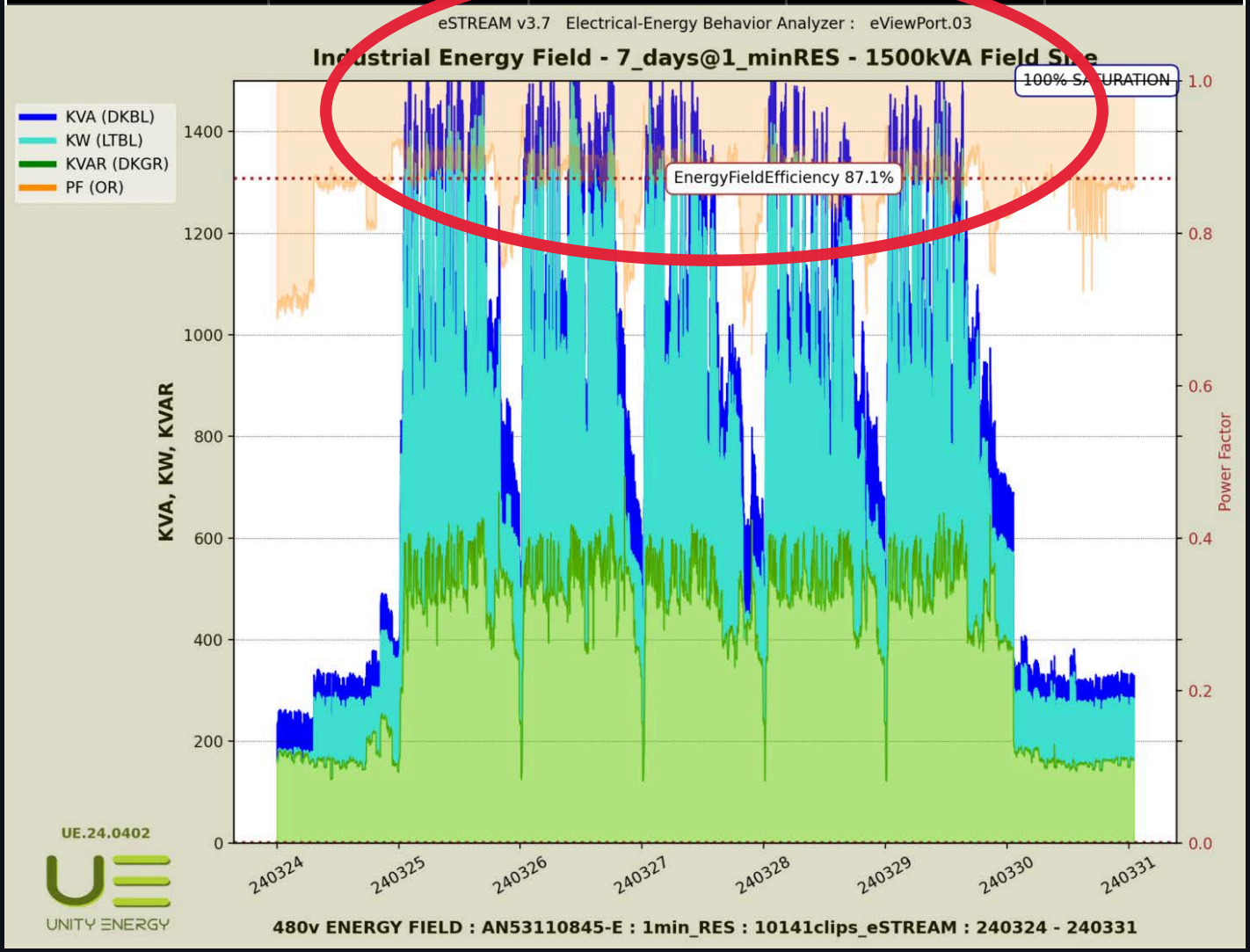
T15 Fillet -- 1500kVA Transformer -- Saturation Review

1 minute Resolution - Showing More Saturation Points

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
90 homes or 231 EVs (1.26kW/hr) (powered continuously) (.35kW/hr)	1512.2 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reaction losses)	RECOVERABLE CO2e
109.4 kVA (unconsumed power)	851.1 kVA 612,810.1 kVA /mo	741.8 kW 534,063.6 kW /mo	401.6 kVAR 289,180.6 kVAR /mo	94.1 LBS 373.7 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
473.8 VOLTS 495.0 MAX 417.2 MIN	1023.7 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	-13.0 % LIMIT [illegible] kVA MAX	87.1 % PF (power factor)



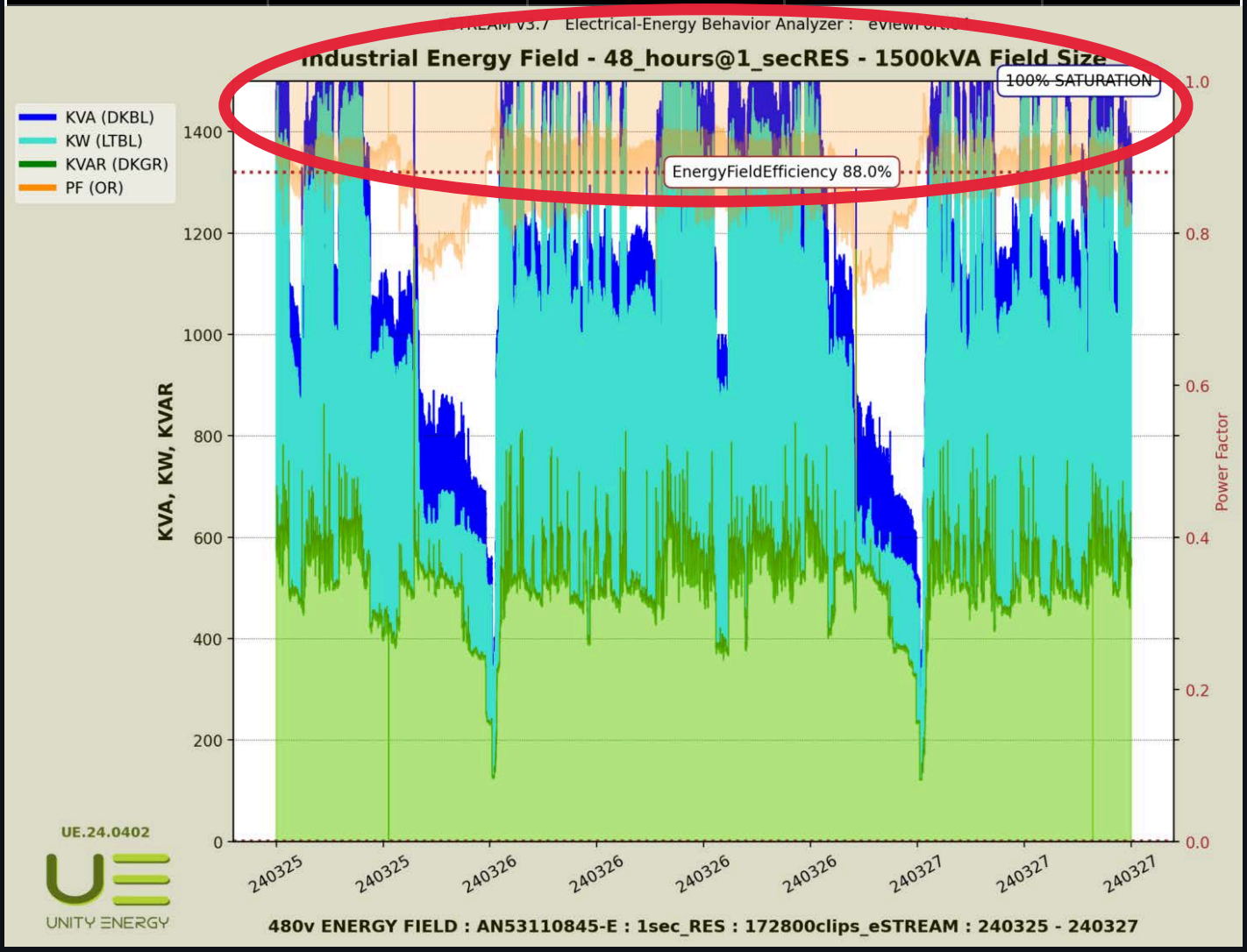
T15 Fillet -- 1500kVA Transformer -- Saturation Review

1 minute Resolution - Showing More Saturation Points

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
110 homes or 281 EVs (1.26kW/hr) (powered con nuously) (.35kW/hr)	1681.8 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reac on losses)	RECOVERABLE CO2e
133.3 kVA (unconsumed power)	1.1 MVA 800,274.9 kVA /mo	978.2 kW 704,286.8 kW /mo	507.8 kVAR 365,597.4 kVAR /mo	114.7 LBS 455.6 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
465.6 VOLTS 477.3 MAX 0.0 MIN	1336.9 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	-26.2 % LIMIT 1845.4 kVA MAX	88.0 % PF (power factor)



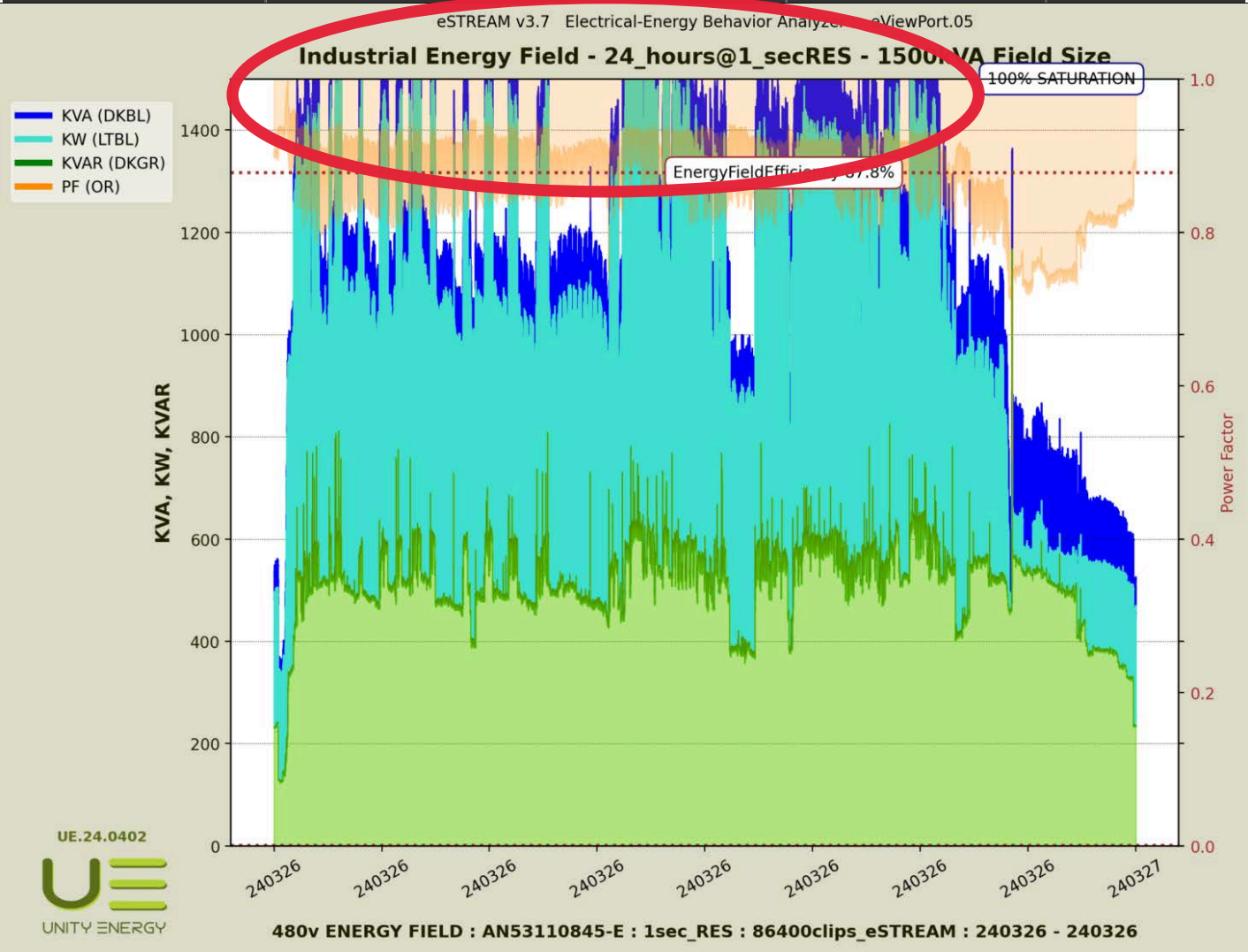
T15 Fillet -- 1500kVA Transformer -- Saturation Review

1secHI-Resolution - FULL KW Saturation

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
112 homes or 287 EVs (1.26kW/hr) (powered continuously) (.35kW/hr)	1681.8 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reaction losses)	RECOVERABLE CO2e
136.1 kVA (unconsumed power)	1.1 MVA 800,310.3 kVA /mo	975.4 kW 702,307.5 kW /mo	508.8 kVAR 366,312.5 kVAR /mo	117.1 LBS 465.1 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
464.7 VOLTS 477.2 MAX 450.6 MIN	1337.0 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	-26.2 % LIMIT 1845.4 kVA MAX	87.8 % PF (power factor)

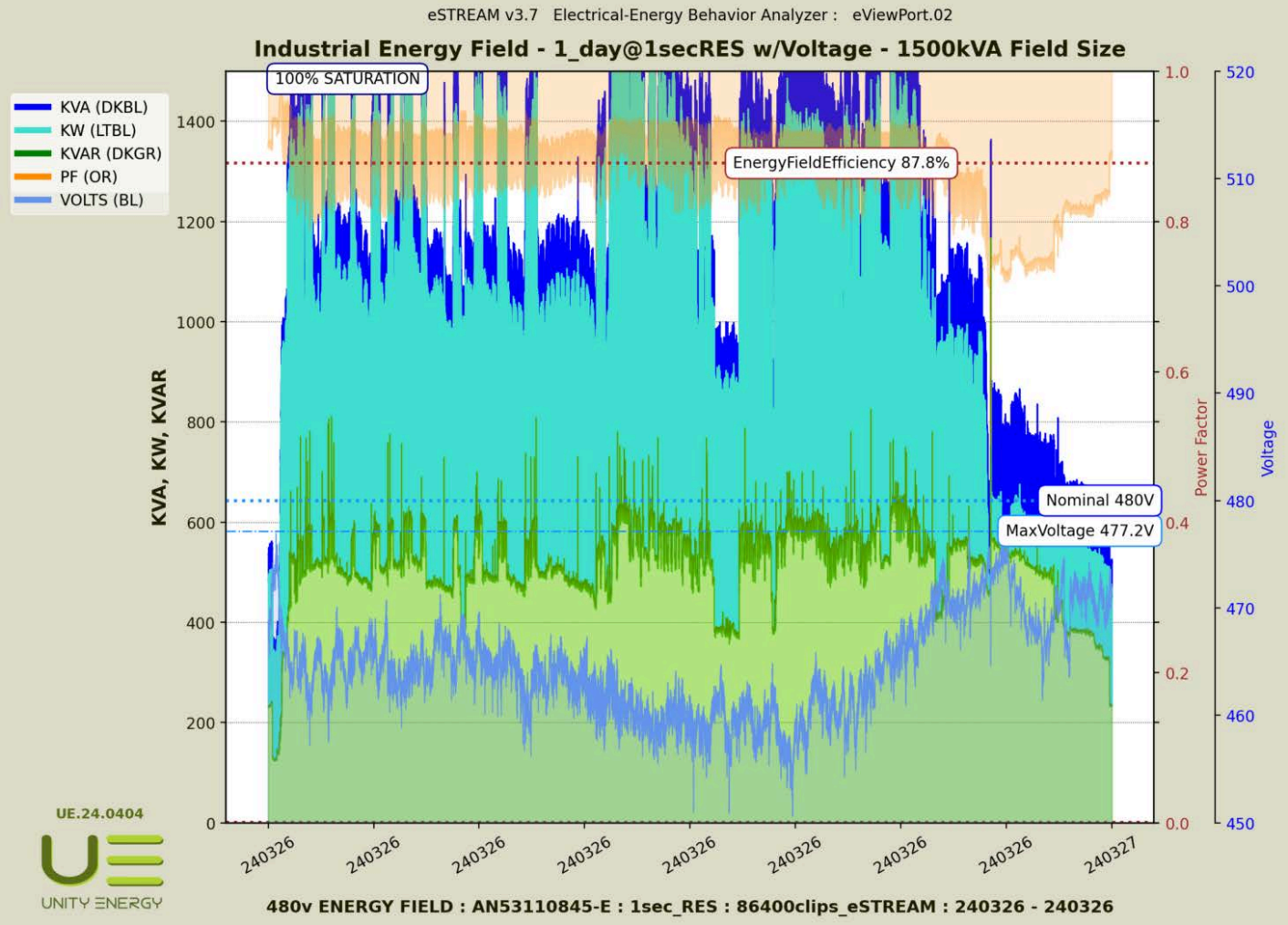


T15 Fillet -- 1500kVA Transformer -- Saturation Review
1secHI-Resolution - FULL KW Saturation w/Voltage

UNRECOVERED ENERGY EQUIVALENT	PEAK DEMAND	Hourly Data UOS
112 homes or 287 EVs (1.26kW/hr) (powered continuously) (.35kW/hr)	1681.8 kW (per this period)	RELATIVE VALUES

RECOVERABLE ENERGY	APPARENT (delivered power)	ACTIVE (consumed power)	REACTIVE (reaction losses)	RECOVERABLE CO2e
136.1 kVA (unconsumed power)	1.1 MVA 800,310.3 kVA /mo	975.4 kW 702,307.5 kW /mo	508.8 kVAR 366,312.5 kVAR /mo	117.1 LBS 465.1 MT /year

VOLTAGE	AMPERAGE	HARMONICS	XFMR CAPACITY	POWER FACTOR
464.7 VOLTS 477.2 MAX 450.6 MIN	1337.0 AMPS (electric current)	0.0 % iTHD 0.0 % iTHD MAX	-26.2 % LIMIT 1845.4 kVA MAX	87.8 % PF (power factor)



Understanding the Dangers of Prolonged Saturation in Large Industrial Class 480v Transformers

Large Class 480v transformers are crucial components in the distribution of power within industrial settings. These transformers come in two main configurations - oil-immersed and dry-type and are responsible for transferring electromagnetic energy efficiently. However, they are prone to a dangerous condition called saturation. Saturation significantly affects the transformer's ability to function within its designed parameters. When a transformer's core becomes saturated, it can cause excessive heat generation, increased energy consumption, and a decrease in the transformer's lifespan. Oil-immersed transformers are at greater risk of saturation, which can speed up the degradation of the insulating oil, leading to insulation breakdown and catastrophic failures such as fires or explosions. Dry-type transformers also face problems under saturation conditions, which can cause insulation failure, leading to arcs or short circuits with disastrous effects on the transformer and the surrounding infrastructure.

The dangers of prolonged undetected saturation in these transformers are serious. From a financial perspective, the loss in efficiency and increased energy consumption can lead to substantial costs for industrial operations. Furthermore, the potential for catastrophic failures poses severe safety risks to personnel and can result in significant operational downtimes, further increasing financial losses.

Preventative measures, such as regular monitoring of transformer performance, installation of protective devices, and adherence to maintenance schedules, are essential in reducing these risks. Advanced diagnostic tools and technologies, such as online monitoring systems, can be instrumental in detecting early signs of saturation, allowing for timely interventions before irreversible damage occurs.

In summary, the risk of prolonged undetected saturation in large Industrial Class 480v transformers is a significant issue for industrial operations. Both oil and dry-type transformers are susceptible to the harmful effects of saturation, which can result in operational inefficiencies, safety hazards, and substantial financial losses. Industries that rely on these transformers need to be proactive in their approach to maintenance and monitoring, using advanced technologies to detect and reduce the risks associated with saturation. By paying careful attention to these aspects, the industrial sector can protect its operations from the hidden dangers of transformer saturation, ensuring its electrical infrastructure's continuity, safety, and efficiency.

CONTACT DETAILS

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After almost 200 years, science has finally allowed humans to quietly develop a proven technology class of Products that have become the cornerstone of Unity Systems.

Unity Energy offers the first complete Industrial Power Management System that continuously optimizes each 480v Energy field, approaching near-perfect UNITY without using any auxiliary conditioning power.

Our system dynamically converts the unconsumed reactive energy to consumable power at the end loads. This behavior can reduce the amount of Utility Energy required to create the Facility Power by almost 30%.

Our system also raises the intellectual awareness of this critical commodity, enabling companies to become more accountable and better stewards of the Energy we consume for our Electricity needs.

Welcome to the Future of Industrial Electricity

UNITY ENERGY with patented MPTS Technology

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